Residual Estimator and Reliability

Part 2: independent study

Local approximation estimates and an energy-error bound

Reading outline

Continue after the quasi-interpolation construction in Part 1.

1. Identify the element and facet patches and use the two local estimates.
2. Derive the mesh-size weights, define η_e and η, and calculate a two-triangle example.
3. Combine Cauchy–Schwarz and finite patch overlap to prove reliability.
4. Return to the nonuniform 1D example and compare indicators with errors.

The local estimates for $\mathcal{Q}_h$ are stated as a hypothesis, not proved here. The remaining slides show how that hypothesis yields the error bound.

Starting identities from Part 1

Use the same diffusion form and homogeneous Dirichlet test spaces as in Part 1:

$$a(w,v)=\int_\Omega k\nabla w\cdot\nabla v\,\mathrm{d}x,\quad \|v\|_a=\sqrt{a(v,v)},\quad \mathcal{V}_{h,0}\subset\mathcal{V}_0.$$

Here $\ell(v)=\int_\Omega fv\,\mathrm{d}x+\int_{\Gamma_N}qv\,\mathrm{d}s$ and $R(u_h;v)=\ell(v)-a(u_h,v)$. With $e=u-u_h$,

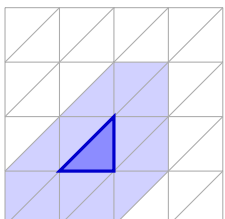
$$R(u_h;v)=a(e,v),\qquad R(u_h;v_h)=0\quad(v_h\in\mathcal{V}_{h,0}).$$

The averaging operator maps $\mathcal{Q}_h:\mathcal{V}_0\to\mathcal{V}_{h,0}$. Taking $v=e$ and $v_h=\mathcal{Q}_he$ gives the identity used below:

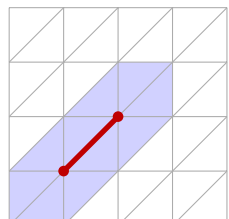
$$\boxed{\|e\|_a^2=R(u_h;e-\mathcal{Q}_he).}$$

The patches in the local estimates

An **element patch** ω_e contains all elements sharing a vertex with $\mathcal{K}_e$. A **facet patch** $\omega_{\mathcal{F}}$ contains all elements sharing a vertex with $\mathcal{F}$.



Dark triangle: $\mathcal{K}_e$
Blue region: ω_e



Red edge: $\mathcal{F}$
Blue region: $\omega_{\mathcal{F}}$

These neighborhoods appear because each coefficient of $\mathcal{Q}_h v$ uses a vertex patch, rather than a value on one element alone.

Assumptions for the residual error bound

The estimates below use the following setting, as in Chapter 8.

- The mesh consists of affine triangles or tetrahedra. It is conforming, boundary fitted, and shape regular. The geometric constants then stay bounded as the mesh is refined.
- The finite element space is the continuous Lagrange space of fixed degree $p \geq 1$. In particular, it contains the piecewise-linear approximation $\mathcal{Q}_h v$.
- The coefficient satisfies $k|_{\mathcal{K}_e} \in C^1(\overline{\mathcal{K}_e})$. It may jump between elements. The element divergence and facet fluxes can therefore be evaluated separately on each element.
- The discrete equations are solved exactly, and the forms are evaluated exactly. These assumptions give $R(u_h; v_h) = 0$ for all $v_h \in \mathcal{V}_{h,0}$.

Local approximation estimates for $\mathcal{Q}_h$

For a shape-regular mesh in $d = 2$ or 3, use the following standard quasi-interpolation estimates for $v \in \mathcal{V}_0$:

$$\boxed{\begin{aligned} \|v - \mathcal{Q}_h v\|_{L^2(\mathcal{K}_e)} &\le C_Q h_e \, |v|_{H^1(\omega_e)}, \\ \|v - \mathcal{Q}_h v\|_{L^2(\mathcal{F})} &\le C_Q h_{\mathcal{F}}^{1/2} \, |v|_{H^1(\omega_{\mathcal{F}})}. \end{aligned}}$$

Here $h_e = \operatorname{diam}(\mathcal{K}_e)$, $h_{\mathcal{F}} = \operatorname{diam}(\mathcal{F})$, and

$$|v|_{H^1(\omega)} = \left(\int_\omega |\nabla v|^2 \, \mathrm{d}x \right)^{1/2}.$$

The constant C_Q does not depend on mesh size. We use these estimates without proving them. They are Hypothesis (Q) in Chapter 8.

Schöberl, §4.3, Theorem 75, p. 65. The element and facet powers are h_e and $h_{\mathcal{F}}^{1/2}$, respectively.

Applying the subtraction identity to the error

The operator gives $\mathcal{Q}_h e \in \mathcal{V}_{h,0}$. Set $v_h = \mathcal{Q}_h e$ in the subtraction identity:

$$\begin{aligned} \|e\|_a^2 &= R(u_h; e) \\ &= R(u_h; e) - \underbrace{R(u_h; \mathcal{Q}_h e)}_{0} \\ &= R(u_h; e - \mathcal{Q}_h e). \end{aligned}$$

Write $w = e - \mathcal{Q}_h e \in \mathcal{V}_0$. Substituting this particular test function into the element-and-facet residual representation gives

$$\boxed{\|e\|_a^2 = \sum_{e'} \int_{K_{e'}} r_{e'} w \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s.}$$

Neither e nor $\mathcal{Q}_h e$ is computed when evaluating the estimator.

The element term produces the weight h_e

Recall $w = e - \mathcal{Q}_h e$. For one element $\mathcal{K}_{e'}$,

$$\left| \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \right| \leq \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, \|w\|_{L^2(\mathcal{K}_{e'})}$$

$$\leq C_Q \qquad \underbrace{h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}}_{\text{computable weighted element residual}} \qquad |e|_{H^1(\omega_{e'})} \, .$$

Use Cauchy–Schwarz on $\mathcal{K}_{e'}$, then the element estimate with $v = e$, for which $v - \mathcal{Q}_h v = w$.
The square of the computable factor is

$$\left(h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \right)^2 = h_{e'}^2 \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 \, .$$

This is the element contribution used in the estimator.

The facet term produces the weight $h_{\mathcal{F}}^{1/2}$

For one interior or Neumann facet,

$$\left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right| \leq \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \, \|w\|_{L^2(\mathcal{F})}$$

$$\leq C_Q \qquad \underbrace{h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}}_{\text{computable weighted facet residual}} \qquad |e|_{H^1(\omega_{\mathcal{F}})} .$$

Use Cauchy–Schwarz on $\mathcal{F}$, then the facet estimate with $v = e$, for which $v - \mathcal{Q}_h v = w$. The square of the computable factor is

$$\left(h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \right)^2 = h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 .$$

This is the facet contribution used in the estimator.

The global residual estimator

Let

$$\mathcal{F}_h^* = \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N.$$

For $d = 2$ or 3, define the **global residual estimator** by collecting the squared computable factors from the two bounds:

$$\boxed{\eta^2 = \sum_e h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \cdot}$$

Every quantity in η is computable from the mesh, the data, and u_h.
The reliability proof will compare η with the unknown energy error.

Local indicators distribute the global estimator

For each element $\mathcal{K}_e$, define the **local error indicator** by

$$\eta_e^2 = h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{\mathrm{int}}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} + \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{N}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \, .$$

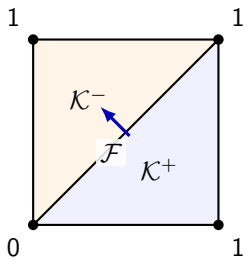
On summing over elements, each interior contribution has total weight $1/2 + 1/2 = 1$. Each Neumann contribution is counted once. Hence

$$\eta^2 = \sum_e \eta_e^2 .$$

The values η_e help select elements for refinement. They are not exact local errors.

Two-triangle example: zero element residuals

Consider $-\Delta u = 0$ on $\Omega = (0,1)^2$, with Dirichlet values shown at the four vertices and varying linearly along each boundary edge. Use the two P_1 triangles shown below.



$$\boldsymbol{n}_{\mathcal{F}} = \frac{(-1,1)}{\sqrt{2}}$$

All vertices lie on the boundary. The unique linear function matching the three nodal values on each triangle is

$$u_h^+(x_1,x_2) = x_1, \qquad u_h^-(x_1,x_2) = x_2.$$

Therefore,

$$\nabla u_h^+ = (1,0), \qquad \nabla u_h^- = (0,1).$$

Because $f = 0$ and these gradients are constant,

$$r_\pm = \nabla\cdot(\nabla u_h^\pm) = 0.$$

The boundary data are imposed exactly, so $e = 0$ on $\partial\Omega$.

Compute the facet term and the local indicators

Using the normal shown on the previous slide,

$$\begin{aligned} j_{\mathcal{F}} &= -\big(\nabla u_h^+ - \nabla u_h^-\big) \cdot \boldsymbol{n}_{\mathcal{F}} \\ &= -\big((1,0) - (0,1)\big) \cdot \frac{(-1,1)}{\sqrt{2}} = \sqrt{2}. \end{aligned}$$

The diagonal has length $h_{\mathcal{F}} = \sqrt{2}$, so

$$\|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = \int_{\mathcal{F}} 2 \, \mathrm{d}s = 2\sqrt{2}, \qquad \eta^2 = h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = 4.$$

The shared-facet contribution is divided equally between its two neighboring elements:

$$\eta_+^2 = \eta_-^2 = 0 + \frac{1}{2}\,(4) = 2, \qquad \eta^2 = \eta_+^2 + \eta_-^2.$$

Since $r_\pm = 0$ and there are no Neumann facets, the residual formula reduces to $R(u_h; v) = \sqrt{2} \int_{\mathcal{F}} v \, \mathrm{d}s$. For example, $v = x_1(1-x_1)x_2(1-x_2)$ is zero on $\partial\Omega$ and positive inside $\mathcal{F}$, so $R(u_h; v) > 0$. Element terms alone miss this residual.

Reliability of the estimator

Combine the weighted residual bounds and obtain a global energy-error estimate

Combining the weighted residual bounds

With $w = e - \mathcal{Q}_h e$ and $\mathcal{F}_h^* = \mathcal{F}_h^{\text{int}} \cup \mathcal{F}_h^N$, start from

$$\begin{aligned}\|e\|_a^2 &= \sum_{e'} \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^*} \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \\ &\leq \sum_{e'} \left| \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \right| + \sum_{\mathcal{F} \in \mathcal{F}_h^*} \left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right|.\end{aligned}$$

Apply the element and facet bounds to these absolute values:

$$\begin{aligned}\|e\|_a^2 \leq C_Q \Bigg[&\sum_{e'} h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, |e|_{H^1(\omega_{e'})} \\ &+ \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \, |e|_{H^1(\omega_{\mathcal{F}})} \Bigg].\end{aligned}$$

The two lists in Cauchy–Schwarz

For nonnegative numbers A_i and B_i, Cauchy–Schwarz gives

$$\sum_i A_i B_i \le \left(\sum_i A_i^2 \right)^{1/2} \left(\sum_i B_i^2 \right)^{1/2}.$$

Use one combined list, with an entry for every element and every facet:

Entry	A_i: weighted residual	B_i: patch seminorm
Element e'	$h_{e'} \, \lVert r_{e'} \rVert_{L^2(\mathcal{K}_{e'})}$	$\lvert e \rvert_{H^1(\omega_{e'})}$
Facet $\mathcal{F}$	$h_{\mathcal{F}}^{1/2} \, \lVert j_{\mathcal{F}} \rVert_{L^2(\mathcal{F})}$	$\lvert e \rvert_{H^1(\omega_{\mathcal{F}})}$

The sum $\sum_i A_i B_i$ is exactly the bracket in the preceding bound.

Applying Cauchy–Schwarz to both residual sums

Substitute the entries in the table into $\|e\|_a^2 \le C_Q \sum_i A_i B_i$ and apply Cauchy–Schwarz:

$$\|e\|_a^2 \le C_Q \underbrace{\left(\sum_{e'} h_{e'}^2 \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 + \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 \right)^{1/2}}_{\eta} \times \left(\sum_{e'} |e|_{H^1(\omega_{e'})}^2 + \sum_{\mathcal{F} \in \mathcal{F}_h^*} |e|_{H^1(\omega_{\mathcal{F}})}^2 \right)^{1/2}.$$

The first factor is exactly the estimator already defined. The remaining task is to bound the sum of the patch seminorms.

Finite patch overlap gives a global error norm

Let $m_{e'}$ be the number of element and facet patches containing $\mathcal{K}_{e'}$. Shape regularity gives a mesh-independent bound $m_{e'} \leq N_{\mathrm{ov}}$. Expand each squared seminorm as an integral:

$$\begin{aligned}
&\sum_{e'} |e|^2_{H^1(\omega_{e'})} + \sum_{\mathcal{F}\in\mathcal{F}_h^*} |e|^2_{H^1(\omega_{\mathcal{F}})} \\
&= \sum_{e'} m_{e'} \int_{\mathcal{K}_{e'}} |\nabla e|^2 \, \mathrm{d}x \\
&\leq N_{\mathrm{ov}} \sum_{e'} \int_{\mathcal{K}_{e'}} |\nabla e|^2 \, \mathrm{d}x = N_{\mathrm{ov}} \, |e|^2_{H^1(\Omega)} \, .
\end{aligned}$$

Thus each element integral is counted at most N_{ov} times. Take the square root and substitute into the second Cauchy–Schwarz factor:

$$\|e\|_a^2 \leq C_Q \sqrt{N_{\mathrm{ov}}} \, \eta \, |e|_{H^1(\Omega)} \, .$$

Reliability of the residual estimator

Since $k(\boldsymbol{x}) \geq k_0 > 0$, the H^1 seminorm satisfies

$$|e|^2_{H^1(\Omega)} = \int_\Omega |\nabla e|^2 \, \mathrm{d}x \leq \frac{1}{k_0} \int_\Omega k|\nabla e|^2 \, \mathrm{d}x = \frac{1}{k_0} \|e\|^2_a .$$

In $\|e\|^2_a \leq C_Q \sqrt{N_{\mathrm{ov}}}\, \eta \, |e|_{H^1(\Omega)}$, replace the seminorm by $\|e\|_a / \sqrt{k_0}$:

$$\|e\|^2_a \leq C_Q \sqrt{\frac{N_{\mathrm{ov}}}{k_0}}\, \eta \, \|e\|_a .$$

If $e \neq 0$, divide by $\|e\|_a$. If $e = 0$, the bound is immediate:

$$\boxed{\|u - u_h\|_a \leq C_Q \sqrt{\frac{N_{\mathrm{ov}}}{k_0}}\, \eta.}$$

This is the reliability bound, with $C_{\mathrm{rel}} = C_Q \sqrt{N_{\mathrm{ov}}/k_0}$ independent of mesh size.

What reliability tells us

The result just proved is

$$\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta.$$

On a shape-regular mesh family, C_{rel} stays fixed as the mesh is refined. Consequently,

$$\eta \longrightarrow 0 \quad \Longrightarrow \quad \|u - u_h\|_a \longrightarrow 0.$$

The computed η is *not itself* a guaranteed numerical upper bound: the theorem contains the constant C_{rel}, which we have not computed.
Likewise, η_e is a contribution to the estimator, not the exact error on $\mathcal{K}_e$. Reliability is a global statement. A separate estimate is needed to compare η_e with the error on its surrounding patch.

Recall the nonuniform 1D example

The problem and computed nodal values from Part 1 are

$$-u'' = 1 \text{ on } (0,1), \qquad u(0) = 0, \quad u'(1) = 0,$$

$$(x_1, x_2, x_3) = (0, 1/3, 1), \qquad (U_1, U_2, U_3) = (0, 5/18, 1/2).$$

The element lengths and slopes are

$$(h_1, h_2) = (1/3, 2/3), \qquad (u_h'|_{\mathcal{K}_1}, u_h'|_{\mathcal{K}_2}) = (5/6, 1/3).$$

Thus $r_1 = r_2 = 1$, the interior nodal residual is $j_2 = -1/2$, and the Neumann residual is $j_3 = -1/3$. For $v(0) = 0$,

$$R(u_h; v) = \int_{\mathcal{K}_1} v \, \mathrm{d}x + \int_{\mathcal{K}_2} v \, \mathrm{d}x - \tfrac{1}{2} v(x_2) - \tfrac{1}{3} v(x_3).$$

Returning to 1D: the nodal terms in the error bound

On an interval, $e \in H^1(0,1)$ has nodal values. We can choose $v_h = \mathcal{I}_h e \in \mathcal{V}_{h,0}$ and set $w = e - \mathcal{I}_h e$. At each node, $w(x_i) = e(x_i) - (\mathcal{I}_h e)(x_i) = 0$.
Start from the subtraction identity, then use the 1D residual formula:

$$\begin{aligned}
\|e\|_a^2 &= R(u_h; e) - \underbrace{R(u_h; \mathcal{I}_h e)}_{0} = R(u_h; w) \\
&= \sum_{e'=1}^{2} \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \underbrace{-\frac{1}{2} w(x_2) - \frac{1}{3} w(x_3)}_{0} \\
&\leq \sum_{e'=1}^{2} \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \|w\|_{L^2(\mathcal{K}_{e'})} .
\end{aligned}$$

The last step uses Cauchy–Schwarz on each element. The nodal residuals are nonzero, but their products with $w(x_i)$ vanish.

The 1D interpolation bound and Cauchy–Schwarz

For linear nodal interpolation on an interval, use the estimate

$$\|v - \mathcal{I}_h v\|_{L^2(\mathcal{K}_{e'})} \le h_{e'} \, \|\mathrm{d}v/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})}, \qquad v \in H^1(0,1).$$

Use this estimate from Chapter 8 without proof. Set $v = e$, so $v - \mathcal{I}_h v = w$, then apply Cauchy–Schwarz to the sum:

$$\begin{aligned} \|e\|_a^2 &\le \sum_{e'=1}^{2} h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, \|\mathrm{d}e/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})} \\ &\le \underbrace{\left(\sum_{e'=1}^{2} h_{e'}^2 \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 \right)^{1/2}}_{\eta_{1\mathrm{D}}} \left(\sum_{e'=1}^{2} \|\mathrm{d}e/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})}^2 \right)^{1/2}. \end{aligned}$$

Since the elements partition $(0,1)$ and $k = 1$, the last factor is $\|\mathrm{d}e/\mathrm{d}x\|_{L^2(0,1)} = \|e\|_a$. Cancel $\|e\|_a$ when nonzero to obtain $\|e\|_a \le \eta_{1\mathrm{D}}$. For $e = 0$, this bound is immediate.

The actual element errors in the 1D example

The exact solution is $u(x) = x - x^2/2$: it satisfies $-u'' = 1$, $u(0) = 0$, and $u'(1) = 0$. Thus $u' = 1 - x$. Subtract the two computed slopes:

$$(u - u_h)' = \begin{cases} (1-x) - 5/6 = 1/6 - x, & x \in \mathcal{K}_1 = (0, 1/3), \\ (1-x) - 1/3 = 2/3 - x, & x \in \mathcal{K}_2 = (1/3, 1). \end{cases}$$

Since $k = 1$, integrate the square to obtain each local energy error:

$$\int_{\mathcal{K}_1} |u' - u_h'|^2 \, \mathrm{d}x = \int_0^{1/3} (x - 1/6)^2 \, \mathrm{d}x = \left[\frac{(x-1/6)^3}{3} \right]_0^{1/3} = \frac{1}{324},$$

$$\int_{\mathcal{K}_2} |u' - u_h'|^2 \, \mathrm{d}x = \int_{1/3}^{1} (x - 2/3)^2 \, \mathrm{d}x = \left[\frac{(x-2/3)^3}{3} \right]_{1/3}^{1} = \frac{8}{324}.$$

Element size in the same 1D example

Recall $h_1 = 1/3$, $h_2 = 2/3$, and $r_1 = r_2 = 1$. For each element, the squared contribution to $\eta_{1\mathrm{D}}$ is

$$h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} = h_e^2 \int_{\mathcal{K}_e} 1 \, \mathrm{d}x = h_e^2 h_e = h_e^3.$$

Compare these contributions with the squared errors just calculated:

Element	h_e	h_e^3	$\displaystyle\int_{\mathcal{K}_e} \lvert u' - u_h' \rvert^2 \, \mathrm{d}x$
$\mathcal{K}_1$	$1/3$	$1/27$	$1/324$
$\mathcal{K}_2$	$2/3$	$8/27$	$8/324$

Here $h_2 = 2h_1$, so $h_2^3/h_1^3 = 2^3 = 8$. The squared local errors also have ratio 8. The residuals are equal, but element size makes the longer element contribute more to both the estimator and the error.

Summary and further reading

- $R(u_h; v) = a(u - u_h, v)$ connects the computable residual functional to the unknown error.
- Elementwise integration by parts separates balance defects inside elements from flux defects on interior and Neumann facets.
- Galerkin orthogonality gives $R(u_h; v) = R(u_h; v - \mathcal{Q}_h v)$. Local approximation estimates then supply the powers of the mesh size.
- The indicators η_e identify where the residual is concentrated, while reliability bounds the global energy error by $C_{\mathrm{rel}}\eta$.

Exit question. For P_1 elements with elementwise-constant k and $f = 0$, can η still be nonzero? Identify the residual term that remains.

Part 3 reading. Interpret reliability, efficiency, and effectivity, then use the indicators in a solve–estimate–mark–refine loop.

References for the construction and estimates

- J. Schöberl, *Numerical Methods for Partial Differential Equations*, §4.3, pp. 64–67. Definition 74 gives the patch-average construction. Theorems 75–76 give its local estimates and reliability. Online version: §27.1–27.2.
- D. N. Arnold, *Lecture Notes on Numerical Analysis of Partial Differential Equations*, Chapter 4, §9.1–9.4, pp. 67–73. Equations (4.16)–(4.18) give local estimates for a projection-based Clément operator. Equations (4.19)–(4.21) derive the residual bound.
- M. G. Larson and F. Bengzon, *The Finite Element Method: Theory, Implementation, and Practice*, §4.9.1, pp. 88–89, Theorem 4.7. See also §2.6.1, pp. 38–39, for the 1D argument.

The references develop the homogeneous Dirichlet case. Our proof retains the Neumann residual and uses the boundary-preserving estimates stated in Chapter 8 as Hypothesis (Q).